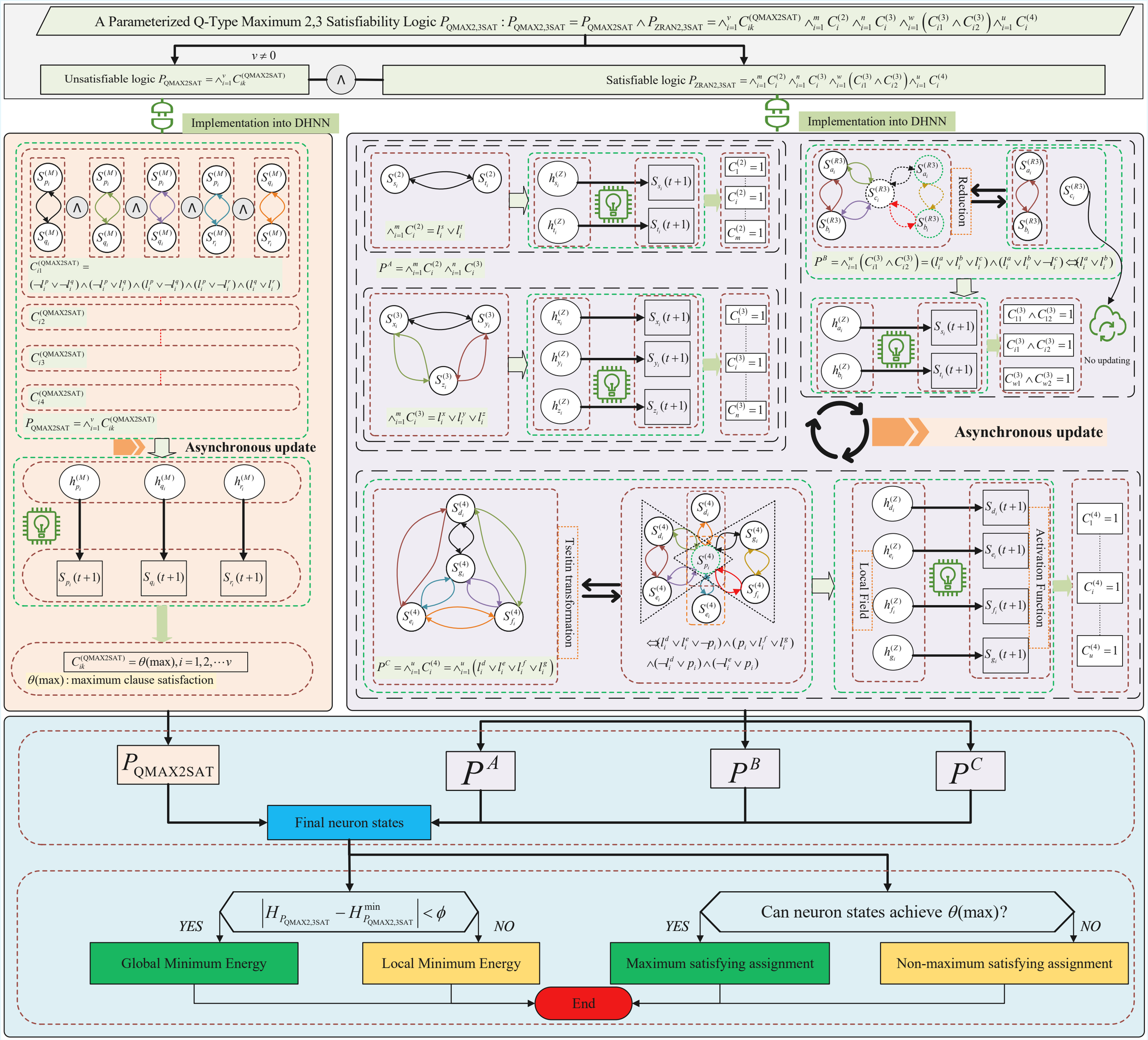
A Parameterized Q-Type Maximum 2,3 Satisfiability Logic $P_{\text{QMAX2,3SAT}}$: $P_{\text{QMAX2,3SAT}} = P_{\text{QMAX2SAT}} \wedge P_{\text{ZRAN2,3SAT}} = \wedge_{i=1}^{v} C_{ik}^{(\text{QMAX2SAT})} \wedge_{i=1}^{m} C_i^{(2)} \wedge_{i=1}^{n} C_i^{(3)} \wedge_{i=1}^{w} \left(C_{i1}^{(3)} \wedge C_{i2}^{(3)}\right) \wedge_{i=1}^{u} C_i^{(4)}$
$v \neq 0$
Unsatisfiable logic $P_{\text{QMAX2SAT}} = \wedge_{i=1}^{v} C_{ik}^{(\text{QMAX2SAT})}$
$\wedge$
Satisfiable logic $P_{\text{ZRAN2,3SAT}} = \wedge_{i=1}^{m} C_i^{(2)} \wedge_{i=1}^{n} C_i^{(3)} \wedge_{i=1}^{w} \left(C_{i1}^{(3)} \wedge C_{i2}^{(3)}\right) \wedge_{i=1}^{u} C_i^{(4)}$
Implementation into DHNN
$C_{i1}^{(\text{QMAX2SAT})} = (\neg l_i^p \vee \neg l_i^q) \wedge (\neg l_i^p \vee l_i^q) \wedge (l_i^p \vee \neg l_i^q) \wedge (l_i^p \vee \neg l_i^r) \wedge (l_i^q \vee l_i^r)$
$C_{i2}^{(\text{QMAX2SAT})}$
$C_{i3}^{(\text{QMAX2SAT})}$
$C_{i4}^{(\text{QMAX2SAT})}$
$P_{\text{QMAX2SAT}} = \wedge_{i=1}^{v} C_{ik}^{(\text{QMAX2SAT})}$
Asynchronous update
$C_{ik}^{(\text{QMAX2SAT})} = \theta(\max), i = 1, 2, \cdots v$
$\theta(\max)$: maximum clause satisfaction
Implementation into DHNN
$\wedge_{i=1}^{m} C_i^{(2)} = l_i^s \vee l_i^t$
$P^A = \wedge_{i=1}^{m} C_i^{(2)} \wedge_{i=1}^{n} C_i^{(3)}$
$\wedge_{i=1}^{m} C_i^{(3)} = l_i^x \vee l_i^y \vee l_i^z$
Reduction
$P^B = \wedge_{i=1}^{w} \left(C_{i1}^{(3)} \wedge C_{i2}^{(3)}\right) = (l_i^a \vee l_i^b \vee l_i^c) \wedge (l_i^a \vee l_i^b \vee \neg l_i^c) \Leftrightarrow (l_i^a \vee l_i^b)$
No updating
Asynchronous update
Tseitin transformation
$P^C = \wedge_{i=1}^{u} C_i^{(4)} = \wedge_{i=1}^{u} \left(l_i^d \vee l_i^e \vee l_i^f \vee l_i^g\right)$
$\Leftrightarrow (l_i^d \vee l_i^e \vee \neg p_i) \wedge (p_i \vee l_i^f \vee l_i^g) \wedge (\neg l_i^d \vee p_i) \wedge (\neg l_i^e \vee p_i)$
Local Field
Activation Function
P_{QMAX2SAT}
P^A
P^B
P^C
Final neuron states
$\left| H_{P_{\text{QMAX2,3SAT}}} - H_{P_{\text{QMAX2,3SAT}}}^{\min} \right| < \phi$
YES
NO
Global Minimum Energy
Local Minimum Energy
Can neuron states achieve $\theta(\max)$?
YES
NO
Maximum satisfying assignment
Non-maximum satisfying assignment
End